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# Does T carry information beyond the SUPPORT pattern (which weight triples are
nonzero; that part is the symmetric zero-sum rule)?
# In the weight basis T'(i,j,k) has 32 nonzero entries. Rephasing an eigenvector (an
arbitrary choice) multiplies entries by phases.
# Gauge-invariant content = products of entries over integer combinations c with
sum_e c_e*(incidence) = 0.  Inputs: T and derived weights only.
import numpy as np, io, contextlib, sympy as sp
with contextlib.redirect_stdout(io.StringIO()):
exec(open('26_roots_from_form_only.py').read())
rng=np.random.RandomState(3); co=rng.randn(len(Hs)); V=[]
for s in range(3):
    Hg=sum(c*H[s] for c,H in zip(co,Hs)); w,VV=np.linalg.eig(Hg);
V.append(VV/np.linalg.norm(VV,axis=0))
Tw=np.einsum('abc,ai,bj,ck->ijk',T,V[0],V[1],V[2]); idx=[tuple(x) for x in
np.argwhere(np.abs(Tw)>1e-9)]
print("nonzero entries:",len(idx)," moduli:",sorted(set(np.round([abs(Tw[e]) for e in
idx],6))))
M=np.zeros((len(idx),24),int)
for r,(i,j,k) in enumerate(idx): M[r,i]=1; M[r,8+j]=1; M[r,16+k]=1
rk=np.linalg.matrix_rank(M); print("gauge parameters 24, rank of their action on the
entries:",rk," -> independent gauge-invariant quantities:",len(idx)-rk)
N=sp.Matrix(M.T.tolist()).nullspace(); vecs=[]
for v in N:
    den=sp.ilcm(*[x.q for x in v]); vecs.append([int(x*den) for x in v])
inv=[]
for c in vecs:
    z=1+0j
    for e,ce in zip(idx,c): z*=Tw[e]**ce
    inv.append(z)
print("gauge-invariant products (modulus, phase/pi):")
ph=[]
for z in inv:
    print("   |z|=%.6f  phase/pi=%.6f"%(abs(z),np.angle(z)/np.pi));
ph.append(round(float(np.angle(z)/np.pi),6))
print("distinct phases (units of pi):",sorted(set(ph)))
print("all invariants real and positive (i.e. every phase removable by
rephasing):",all(abs(z.imag)<1e-9 and z.real>0 for z in inv))
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